

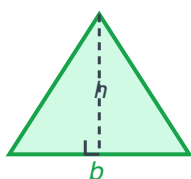
4.2 Mensuration: Area, Perimeter and Volume

OCR GCSE Mathematics J560 | Geometry and Measures | Papers 1, 2 & 3

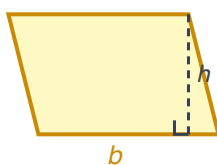
4.2.1 Perimeter and Area of Standard Shapes



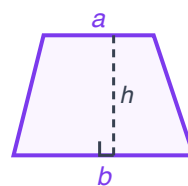
Rectangle



Triangle



Parallelogram



Trapezium

Shape	Area Formula	Perimeter
Rectangle	$A = lw$	$P = 2(l + w)$
Triangle	$A = \frac{1}{2}bh$	Sum of three sides
Parallelogram	$A = bh$ (perpendicular height)	$P = 2(a + b)$
Trapezium	$A = \frac{1}{2}(a + b)h$	Sum of four sides

Exam Tip: For the triangle and parallelogram, h must be the **perpendicular** height — the vertical distance between the parallel sides, not the slant side.

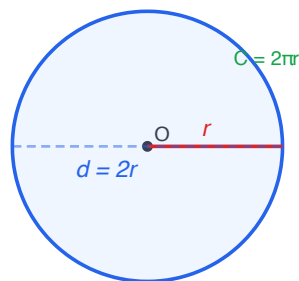
Worked Example: Find the area of a trapezium with parallel sides 8 cm and 14 cm and perpendicular height 5 cm.

$$A = \frac{1}{2}(8 + 14) \times 5 = \frac{1}{2} \times 22 \times 5 = \mathbf{55 \text{ cm}^2}$$

Worked Example: A triangle has base 9 cm and area 27 cm². Find the perpendicular height.

$$27 = \frac{1}{2} \times 9 \times h \Rightarrow h = 27 \times 2 \div 9 = \mathbf{6 \text{ cm}}$$

4.2.2 Area and Circumference of Circles



$$\text{Circumference: } C = 2\pi r = \pi d$$

$$\text{Area: } A = \pi r^2$$

$$\text{For an arc of angle } \theta^\circ: \text{Arc length} = \frac{\theta}{360} \times 2\pi r$$

$$\text{For a sector of angle } \theta^\circ: \text{Sector area} = \frac{\theta}{360} \times \pi r^2$$

Worked Example: Find the circumference and area of a circle with radius 7 cm. (Leave answers in terms of π .)

$$C = 2\pi \times 7 = 14\pi \approx \mathbf{44.0 \text{ cm}}$$

$$A = \pi \times 7^2 = 49\pi \approx \mathbf{153.9 \text{ cm}^2}$$

Worked Example: A sector has radius 12 cm and angle 150° . Find its arc length and area.

$$\text{Arc length} = \frac{150}{360} \times 2\pi \times 12 = \frac{5}{12} \times 24\pi = 10\pi \approx \mathbf{31.4 \text{ cm}}$$

$$\text{Sector area} = \frac{150}{360} \times \pi \times 12^2 = \frac{5}{12} \times 144\pi = 60\pi \approx \mathbf{188.5 \text{ cm}^2}$$

Exam Tip: If the question says "leave in terms of π ", do not press the π button on your calculator — just write 49π etc. If it asks for a decimal, use $\pi \approx 3.14159 \dots$

4.2.3 Area of Composite Shapes

A **composite shape** is made up of two or more standard shapes joined together. Split it into rectangles, triangles, circles, etc. and add or subtract areas.

Worked Example: Find the area of an L-shaped figure formed by a rectangle $10 \text{ cm} \times 8 \text{ cm}$ with a $4 \text{ cm} \times 3 \text{ cm}$ rectangle cut from one corner.

$$\text{Area of full rectangle: } 10 \times 8 = 80 \text{ cm}^2$$

$$\text{Area removed: } 4 \times 3 = 12 \text{ cm}^2$$

$$\text{Composite area} = 80 - 12 = \mathbf{68 \text{ cm}^2}$$

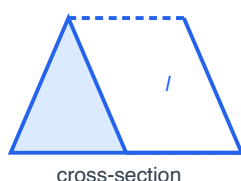
Worked Example: A shape consists of a rectangle $6 \text{ cm} \times 4 \text{ cm}$ with a semicircle of diameter 4 cm attached to one end. Find the total area.

$$\text{Rectangle area} = 6 \times 4 = 24 \text{ cm}^2$$

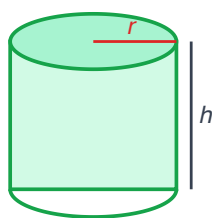
$$\text{Semicircle: radius} = 2 \text{ cm; area} = \frac{1}{2}\pi \times 2^2 = 2\pi \approx 6.28 \text{ cm}^2$$

$$\text{Total} = 24 + 2\pi \approx \mathbf{30.3 \text{ cm}^2}$$

4.2.4 Volume and Surface Area of Prisms and Cylinders



Triangular prism
(length l)



Cylinder

All prisms and cylinders:

$$V = A_{\text{cross-section}} \times l$$

Shape	Volume	Curved Surface Area	Total Surface Area
Prism (general)	$V = A \times l$	Perimeter of cross-section $\times l$	$2A + \text{CSA}$
Cylinder	$V = \pi r^2 h$	$2\pi r h$	$2\pi r h + 2\pi r^2$

Worked Example: Find the volume and total surface area of a cylinder with radius 5 cm and height 12 cm.

Volume: $V = \pi \times 5^2 \times 12 = 300\pi \approx 942.5 \text{ cm}^3$

Curved surface area: $2\pi \times 5 \times 12 = 120\pi \approx 376.99 \text{ cm}^2$

Two circular ends: $2 \times \pi \times 5^2 = 50\pi \approx 157.08 \text{ cm}^2$

Total SA: $120\pi + 50\pi = 170\pi \approx 534.1 \text{ cm}^2$

Worked Example: A triangular prism has a right-angled triangular cross-section with legs 3 cm and 4 cm (hypotenuse 5 cm). The prism has length 10 cm. Find its volume and surface area.

Volume: Cross-section area = $\frac{1}{2} \times 3 \times 4 = 6 \text{ cm}^2$

$$V = 6 \times 10 = \mathbf{60 \text{ cm}^3}$$

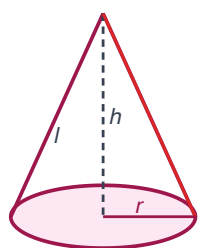
Surface area:

2 triangular ends: $2 \times 6 = 12 \text{ cm}^2$

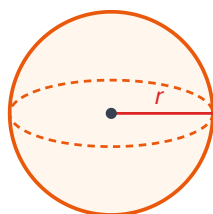
3 rectangular faces: $(3 \times 10) + (4 \times 10) + (5 \times 10) = 30 + 40 + 50 = 120 \text{ cm}^2$

Total SA = $12 + 120 = \mathbf{132 \text{ cm}^2}$

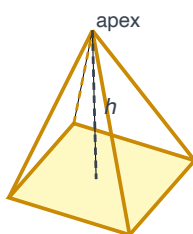
4.2.5 Volume and Surface Area of Pyramids, Cones and Spheres



Cone



Sphere



Square-based pyramid

Shape	Volume	Surface Area
Pyramid (any base)	$V = \frac{1}{3}Ah$	Base area + area of all triangular faces
Cone	$V = \frac{1}{3}\pi r^2 h$	$\pi r l + \pi r^2$ where $l = \sqrt{r^2 + h^2}$
Sphere	$V = \frac{4}{3}\pi r^3$	$4\pi r^2$

l = slant height of cone; h = perpendicular height; A = base area of pyramid.

Worked Example: Find the volume of a cone with radius 6 cm and height 8 cm.

$$V = \frac{1}{3} \times \pi \times 6^2 \times 8 = \frac{1}{3} \times 36\pi \times 8 = 96\pi \approx \mathbf{301.6 \text{ cm}^3}$$

Worked Example: Find the total surface area of a cone with radius 5 cm and height 12 cm.

$$\text{Slant height: } l = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13 \text{ cm}$$

$$\text{Curved SA} = \pi \times 5 \times 13 = 65\pi$$

$$\text{Base area} = \pi \times 5^2 = 25\pi$$

$$\text{Total SA} = 65\pi + 25\pi = 90\pi \approx \mathbf{282.7 \text{ cm}^2}$$

Worked Example: A sphere has volume $288\pi \text{ cm}^3$. Find its radius.

$$\frac{4}{3}\pi r^3 = 288\pi$$

$$r^3 = 288 \times \frac{3}{4} = 216$$

$$r = \sqrt[3]{216} = \mathbf{6 \text{ cm}}$$

Worked Example: A square-based pyramid has base side 9 cm and perpendicular height 10 cm. Find its volume.

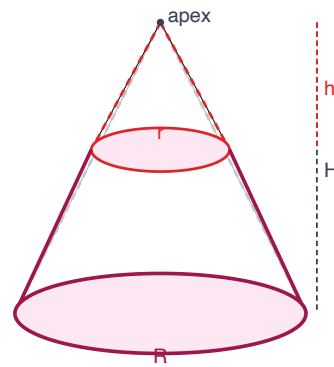
$$\text{Base area } A = 9^2 = 81 \text{ cm}^2$$

$$V = \frac{1}{3} \times 81 \times 10 = \mathbf{270 \text{ cm}^3}$$

4.2.6 Frustums

Higher

A **frustum** is the part of a cone left after the top (a smaller, similar cone) is removed by a cut parallel to the base.



Volume of frustum = Volume of large cone – Volume of small cone:

$$V_{\text{frustum}} = \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi r^2 h = \frac{\pi}{3}(R^2 H - r^2 h)$$

where R = radius of base, r = radius of top, H = height of full cone, h = height of removed small cone.

Exam Tip: To find H and h , use the similar triangle ratio $\frac{r}{R} = \frac{h}{H}$. If you are given only the height of the frustum ($H - h$), you will need to set up a ratio to find H and h separately.

Worked Example: A frustum is formed by removing a small cone of height 6 cm and radius 2 cm from a full cone of height 18 cm and radius 6 cm. Find the volume of the frustum.

Check similarity ratio: $r/R = 2/6 = 1/3$ and $h/H = 6/18 = 1/3$ ✓ (consistent)

$$V_{\text{large}} = \frac{1}{3}\pi \times 6^2 \times 18 = \frac{1}{3}\pi \times 36 \times 18 = 216\pi$$

$$V_{\text{small}} = \frac{1}{3}\pi \times 2^2 \times 6 = \frac{1}{3}\pi \times 4 \times 6 = 8\pi$$

$$V_{\text{frustum}} = 216\pi - 8\pi = 208\pi \approx \mathbf{653.5 \text{ cm}^3}$$

4.2.7 Units of Area and Volume

Units of length: 1 km = 1000 m; 1 m = 100 cm; 1 cm = 10 mm

Units of area:

$$1 \text{ m}^2 = 10,000 \text{ cm}^2 = 100^2 \text{ cm}^2$$

$$1 \text{ km}^2 = 1,000,000 \text{ m}^2$$

$$1 \text{ hectare (ha)} = 10,000 \text{ m}^2$$

Units of volume:

$$1 \text{ m}^3 = 1,000,000 \text{ cm}^3 = 100^3 \text{ cm}^3$$

$$1 \text{ litre} = 1000 \text{ cm}^3 = 1000 \text{ ml}$$

$$1 \text{ m}^3 = 1000 \text{ litres}$$

Conversion	Multiply by	Example
cm^2 to mm^2	$\times 100$	$5 \text{ cm}^2 = 500 \text{ mm}^2$
m^2 to cm^2	$\times 10,000$	$3 \text{ m}^2 = 30,000 \text{ cm}^2$
cm^3 to mm^3	$\times 1000$	$2 \text{ cm}^3 = 2000 \text{ mm}^3$
m^3 to cm^3	$\times 1,000,000$	$0.5 \text{ m}^3 = 500,000 \text{ cm}^3$
cm^3 to litres	$\div 1000$	$5000 \text{ cm}^3 = 5 \text{ litres}$
litres to cm^3	$\times 1000$	$2.5 \text{ litres} = 2500 \text{ cm}^3$

Exam Tip: Remember: when converting units of *area*, square the linear conversion factor. When converting units of *volume*, cube the linear conversion factor. E.g., $1 \text{ m} = 100 \text{ cm}$, so $1 \text{ m}^2 = 100^2 = 10,000 \text{ cm}^2$ and $1 \text{ m}^3 = 100^3 = 1,000,000 \text{ cm}^3$.

Worked Example: A tank has volume 0.36 m^3 . How many litres of water does it hold?

$$0.36 \text{ m}^3 = 0.36 \times 1,000,000 \text{ cm}^3 = 360,000 \text{ cm}^3$$

$$\text{Litres} = 360,000 \div 1000 = \mathbf{360} \text{ litres}$$

Master Formula Summary

Shape	Area / Volume	Perimeter / Surface Area
Rectangle	$A = lw$	$P = 2(l + w)$
Triangle	$A = \frac{1}{2}bh$	Sum of sides
Parallelogram	$A = bh$	$2(a + b)$
Trapezium	$A = \frac{1}{2}(a + b)h$	Sum of sides
Circle	$A = \pi r^2$	$C = 2\pi r$
Sector	$A = \frac{\theta}{360}\pi r^2$	$\frac{\theta}{360} \times 2\pi r + 2r$
Prism	$V = A_{\text{cs}} \times l$	$2A_{\text{cs}} + P_{\text{cs}} \times l$
Cylinder	$V = \pi r^2 h$	$2\pi r h + 2\pi r^2$
Pyramid	$V = \frac{1}{3}Ah$	Base + triangular faces
Cone	$V = \frac{1}{3}\pi r^2 h$	$\pi r l + \pi r^2; l = \sqrt{r^2 + h^2}$
Sphere	$V = \frac{4}{3}\pi r^3$	$SA = 4\pi r^2$
Frustum (Higher)	$V = \frac{\pi}{3}(R^2 H - r^2 h)$	$\pi R l + \pi r l + \pi R^2 + \pi r^2$ (approx.)